Linear Programming and Convex Optimization
Problem Set 3 – CS 6515/4540 (Fall 2026)

This problem set is due on **Tuesday October 13th**. Submission is via Gradescope. Your solution must be a typed pdf (e.g. via LaTeX) – no handwritten solutions.

10 LPs

Alice is trying to get enough oranges and bananas to host a fruit party. To successfully host a party she needs **at least 8 oranges** and **at least 20 bananas**. Unfortunately, her local grocery story only sells fruit in bundles. Bundle A costs 3 dollars and contains one orange and two bananas. Bundle B costs 2 dollars and contains three oranges and a banana. Fortunately, the grocery story will allow Alice to buy fractions of bundles (i.e. she can buy 2.5 bundle As to get 2.5 oranges and 5 bananas). They will not allow Alice to buy negative bundles (i.e. she cannot buy -1 bundle As and 3 bundle Bs to get 5 oranges and a banana).

Alice would like to buy x_A bundle As and x_B bundle Bs to guarantee she has at least 8 oranges and at least 20 bananas. Moreover, she would like to minimize her dollars spent.

1. Write a linear program whose solution is the optimal choice of x_A, x_B for Alice's problem and briefly justify why this is the correct LP.

   We have the following LP:

   $$\begin{aligned} \min \quad & 3x_A + 2x_B \\ s.t.\ & x_A + 3x_B \geq 8 \\ & 2x_A + x_B \geq 20 \\ & x_A, x_B \geq 0 \end{aligned}$$

   In the above, x_A and x_B represent the number of bundle As and bundle Bs to be bought respectively. The objective is to minimize the cost, and since one bundle of A and B cost 3 and 2 respectively, we minimize $3x_A + 2x_B$. The first and second non-trivial constraints represent the constraints that Alice needs to satisfy to host the party. Finally, the trivial constraints ensure that Alice cannot buy negative bundles.

2. Show that there exists a solution to this linear program with objective value 30 (and hence Alice has to spend at most 30 dollars).

   We can verify that $x_A = 10$ and $x_B = 0$ is indeed a feasible solution and leads to objective value 30.

   $$\begin{aligned} & 3x_a + 2x_b = 30 + 0 = 30 \\ & x_A + 3x_B = 10 \geq 8 \\ & 2x_A + x_B = 20 \geq 20 \\ & x_A = 10 \geq 0, x_B = 0 \geq 0 \end{aligned}$$

3. Prove that every feasible solution to this linear program has value at least 30 (prove this by taking a non-negative linear combination of the constraints), so the solution in the previous part is optimal.

We can express the objective as a nonnegative linear combination of constraints and hence obtain a lower bound as follows:

$$\underbrace{3x_A + 2x_B}_{cost} = (3/2) \cdot \underbrace{(2x_A + x_B)}_{\text{constraint } 2x_A + x_B \geq 20} + (1/2) \cdot \underbrace{(x_B)}_{\text{constraint } x_B \geq 0} \geq (3/2) \cdot 20 + (1/2) \cdot 0 = 30$$

11 LP Equivalent Formulation

1. Show that the class of left LPs (LP1) can efficiently represent any LP on the right (LP2). In other words, if we have a polynomial time algorithm to solve any LP1 on the left, then we can solve in polynomial time any LP2 on the right.

$$\begin{aligned} & \max \sum_{i=1}^{n} c_i x_i && \text{(LP1)} \\ \text{s.t.} \quad & \sum_{i=1}^{n} A_{ji} x_i = b_j, \quad \forall j \in \{1, \ldots, m\} \\ & x_i \geq 0, \quad \forall i \in \{1, \ldots, n\}. \end{aligned}$$

$$\begin{aligned} & \max \sum_{i=1}^{n} c_i x_i && \text{(LP2)} \\ \text{s.t.} \quad & \sum_{i=1}^{n} A_{ji} x_i \leq b_j, \quad \forall j \in \{1, \ldots, m\} \end{aligned}$$

(Hint: You need two ideas: (1) We can replace unconstrained x by $x = x^+ - x^-$ where $x^+ \geq 0$ and $x^- \geq 0$. (2) We can replace $Ax \leq b$ by $Ax + Z = b$ for some vector $Z \geq 0$.)

Given LP2, we construct LP1 as

$$\begin{aligned} \max \ & (c^\top, -c^\top, 0) \begin{pmatrix} x^+ \\ x^- \\ z \end{pmatrix} \\ & \big[\mathbf{A} | -\mathbf{A} | \mathbf{I}\big] \begin{pmatrix} x^+ \\ x^- \\ z \end{pmatrix} = b \\ & x^+, x^-, z \geq 0 \\ & x^+, x^- \in \mathbb{R}^n, z \in \mathbb{R}^m \end{aligned}$$

Then solve LP1 and return $x = x^+ - x^-$.

Correctness We now argue that the two LPs are equivalent:
1) For the optimal solution x^+, x^-, z of LP1, define $x = x^+ - x^-$. Then we have $\mathbf{A}x = \mathbf{A}x^+ - \mathbf{A}x^- \leq \mathbf{A}x^+ - \mathbf{A}x^- + z = b$ (where the inequality uses $z \geq 0$). So this is a feasible solution of LP2. It has the same cost as the solution for LP1 because $c^\top x = c^\top x^+ - c^\top x^-$. We next need to argue that no better solution for LP2 exists.
2) Observe that for the optimal solution x of LP2, we can define $x^+ = \max(0, x)$, $x^- = -\min(x, 0)$, $z = b - \mathbf{A}x$. These satisfy $\mathbf{A}x^+ - \mathbf{A}x^- + z = b$ and $c^\top x^+ - c^\top x^- = c^\top x$. Further, $x^+, x^- \geq 0$ by definition and $z \geq 0$ since $\mathbf{A}x \leq b$. So they are a feasible solution for LP1 with the same cost. Thus LP2 cannot have a solution with better cost than the optimal solution of LP1.

2. Now use a similar idea to show that Farkas' Lemma A below implies Farkas' Lemma B.

   - Farkas' Lemma A says that the system of inequalities $\sum_{i=1}^{n} A_{ji} x_i = b_j$ for $j \in \{1, \ldots, m\}$ and

$x_i \geq 0$ for $i \in \{1, \ldots, n\}$ are infeasible iff there exist λ_j for $j \in \{1, \ldots, m\}$ such that

$$\sum_{j=1}^{m} \lambda_j b_j < 0 \qquad \text{and} \qquad \sum_{j=1}^{m} \lambda_j A_{ji} \geq 0 \quad \forall i \in \{1, \ldots, n\}.$$

- Farkas' Lemma B says that the system of inequalities $\sum_{i=1}^{n} A_{ji} x_i \leq b_j$ for $j \in \{1, \ldots, m\}$ are infeasible iff there exist *non-negative* $\lambda_j \geq 0$ for $j \in \{1, \ldots, m\}$ such that

$$\sum_{j=1}^{m} \lambda_j b_j < 0 \qquad \text{and} \qquad \sum_{j=1}^{m} \lambda_j A_{ji} = 0 \quad \forall i \in \{1, \ldots, n\}.$$

Note that $[k]$ is shorthand for the set $\{1, \ldots, k\}$.
We must show both directions of Farkas' Lemma B. We first start with the simpler reverse direction ($\Leftarrow$). So, assume that there exist $\lambda_j \geq 0$ such that:

$$\sum_{j=1}^{m} \lambda_j b_j < 0 \qquad \text{and} \qquad \sum_{j=1}^{m} \lambda_j A_{ji} = 0 \quad \forall i \in \{1, \ldots, n\}.$$

For the sake of contradiction, assume that there exists a feasible solution to the system of inequalities $\sum_{i=1}^{n} A_{ji} x_i \leq b_j$ for every $j \in [m]$, say $\{x^*\}_{i \in [n]}$. Then, for any $j \in [m]$, we have $\sum_{i=1}^{n} A_{ji} x_i^* \leq b_j$. Multiply this inequality by λ_j; notice that we can safely do this while **preserving the direction of the inequality** because $\lambda_j \geq 0$, and so we get $\sum_{i=1}^{n} \lambda_j A_{ji} x_i^* \leq \lambda_j b_j$. Summing all such inequalities for every $j \in [m]$, we get:

$$\sum_{j=1}^{m} \sum_{i=1}^{n} \lambda_j A_{ji} x_i^* \leq \sum_{j=1}^{m} \lambda_j b_j \Rightarrow \sum_{j=1}^{m} \lambda_j b_j \geq \sum_{i=1}^{n} \sum_{j=1}^{m} \lambda_j A_{ji} x_i^* \Rightarrow \sum_{j=1}^{m} \lambda_j b_j \geq 0$$

In the above, we use the fact that for every $i \in [n]$, we have $\sum_{j=1}^{m} \lambda_j A_{ji} = 0$ which implies $\sum_{j=1}^{m} \lambda_j A_{ji} x_i^* = 0$. However, the implication $\sum_{j=1}^{m} \lambda_j b_j \geq 0$ is a contradiction to $\sum_{j=1}^{m} \lambda_j b_j < 0$. Hence, we have the reverse implication.

Now, we prove the forward implication ($\Rightarrow$). We assume that the system of inequalities $\sum_{i=1}^{n} A_{ji} x_i \leq b_j$ is infeasible. Using the trick from Part 1, we can transform this LP to the form of the LP in Farkas Lemma A:

$$\sum_{i=1}^{n} A_{ji} x_i^+ + \sum_{i=1}^{n} (-A_{ji}) x_i^- + z_j = b_j \; \forall \; j \in [m]$$

Now, we have reached the form in Farkas Lemma A, with the matrix "A", the input "x" and vector "b" in the statement of Farkas Lemma A being given by the matrix A', input x' and b' defined as follows (using matrix and vector notation):

$$A' = \begin{pmatrix} \mathbf{A} & -\mathbf{A} & \mathbf{I}_m \end{pmatrix}; x' = \begin{bmatrix} \mathbf{x}^+ \\ \mathbf{x}^- \\ \mathbf{z} \end{bmatrix}; b' = \mathbf{b}$$

Notice that the number of variables (length of x'), and hence the number of columns of A' is equal to $2n + m$, while the number of constraints remains m. Applying Farkas Lemma A, we obtain that there exist $\lambda_j \in \mathbb{R} \; \forall \; j \in [m]$ such that, we have:

$$\sum_{j=1}^{m} \lambda_j A'_{ji} \geq 0 \; \forall \; i \in [2n + m]; \sum_{j=1}^{m} \lambda_j b_j < 0$$

Expanding the condition on the matrix A' above for the last m columns, we effectively take a dot product of each of the last m columns of A, which are given by the identity matrix $\mathbf{I}_m$ with the vector $\{\lambda_j\}_{j \; in [m]}$ giving us $\lambda_j \geq 0$ for every $j \in [m]$. This illustrates that these $\lambda_j \in \mathbb{R}$ are in fact nonnegative, and we also have the condition $\sum_{j=1}^{m} \lambda_j b_j < 0$, and so we have proven one part of Farkas Lemma B.
It remains to show the other part of Farkas Lemma B. Applying the condition on matrix A' to columns i and $i + n$ for every $i \in [n]$, we get:

$$\sum_{j=1}^{m} \lambda_j A_{ji} \geq 0 \text{ and } \sum_{j=1}^{m} \lambda_j (-A_{ji}) \geq 0 \text{ respectively } \Rightarrow \sum_{j=1}^{m} \lambda_j A_{ji} = 0$$

Hence, we have shown that the other condition is also true. This completes the proof.

12 LP for Regression

Suppose we are given n points $(x_1, y_1), (x_2, y_2), ..., (x_n, y_n) \in \mathbb{R} \times \mathbb{R}$. The goal of this problem is to find a line $y = ax + b$, where $a, b \in \mathbb{R}$ that fits these points as closely as possible, where closeness is defined according to some objective function. Write a polynomial-sized LP for the following settings:

1. ℓ_1 regression: Objective is to minimize $\sum_{i=1}^{n} |y_i - ax_i - b|$.

   We introduce variables z_i:

   $$\begin{aligned} \min & \sum_{i=1}^{n} z_i \\ s.t.\ & z_i \geq y_i - ax_i - b \\ & z_i \geq b - y_i + ax_i \\ & z_i \geq 0 \ \forall\ i \in [n], a, b \in \mathbb{R}(\text{unconstrained}) \end{aligned}$$

   Note for any choice of a, b, each fo the z_i's are minimized when they're equal to $\max(y_i - ax_i - b, b - y_i + ax_i) = |y_i - ax_i - b|$, so at the optimum, we are minimizing the correct thing.
   Also note the variables in the LP are the z_i's along with a and b. The only constant terms in these constraints are the y_i's.

2. ℓ_∞ regression: Objective is to minimize $\max_{i=1}^{n} |y_i - ax_i - b|$

   We introduce a single scalar variable Z:

   $$\begin{aligned} \min\ & Z \\ s.t.\ & Z \geq y_i - ax_i - b \\ & Z \geq b - y_i + ax_i \\ & Z \geq 0, a, b \in \mathbb{R}(\text{unconstrained}) \end{aligned}$$

3. Write the dual LPs for both the above LPs.

For ℓ_1: We will associate the variable u_i with the constraints of the form $z_i \geq y_i - ax_i - b$, and v_i with the constraints of the form $z_i \geq b - y_i + ax_i$, giving us $2i$ variables total.
We get a dual:

$$\begin{aligned}
\max \ & \sum_{i=1}^{n} y_i(u_i - v_i) \\
s.t. \ & u_i + v_i \leq 1 && (n \text{ constraints assicated with } z_i\text{'s}) \\
& \sum_{i=1}^{n}(u_i - v_i) = 0 && (\text{constraint assicated with } b) \\
& \sum_{i=1}^{n} x_i(u_i - v_i) = 0 && (\text{constraint assicated with } a) \\
& u_i, v_i \geq 0
\end{aligned}$$

For ℓ_∞: We associate u_i and v_i the same way.
We get a dual:

$$\begin{aligned}
\max \ & \sum_{i=1}^{n} y_i(u_i - v_i) \\
s.t. \ & \sum_{i=1}^{n}(u_i + v_i) \leq 1 && (\text{constraint assicated with } Z) \\
& \sum_{i=1}^{n}(u_i - v_i) = 0 && (\text{constraint assicated with } b) \\
& \sum_{i=1}^{n} x_i(u_i - v_i) = 0 && (\text{constraint assicated with } a) \\
& u_i, v_i \geq 0
\end{aligned}$$

13 The Minimax Theorem via LP Duality

Consider a two-player game specified by a matrix $A \in \mathbb{R}^{m \times n}$, where $m, n \geq 1$. Alice chooses a row $i \in \{1, \ldots, m\}$ and Bob chooses a column $j \in \{1, \ldots, n\}$, simultaneously. Alice receives payoff A_{ij} and Bob receives payoff $-A_{ij}$. Thus Alice wants to maximize Alice's payoff, while Bob wants to minimize it. Entries of A may be positive, zero, or negative.

The players may randomize their choices independently. A *mixed strategy* for Alice is a probability vector $p \in \Delta_m$, where

$$\Delta_k = \left\{ z \in \mathbb{R}^k : z_\ell \geq 0 \text{ for all } \ell, \ \sum_{\ell=1}^{k} z_\ell = 1 \right\}.$$

Similarly, Bob's mixed strategy is $q \in \Delta_n$. Their expected payoff to Alice is

$$p^\top A q = \sum_{i=1}^{m} \sum_{j=1}^{n} p_i A_{ij} q_j.$$

Alice's largest guaranteed expected payoff is $\max_{p \in \Delta_m} \min_{q \in \Delta_n} p^\top A q$. Bob's smallest guaranteed upper bound on Alice's expected payoff is $\min_{q \in \Delta_n} \max_{p \in \Delta_m} p^\top A q$. These guarantees allow an opponent to know the chosen probability vector, but not the outcome of the random choice.

Minimax theorem. For every real matrix A,

$$\boxed{\max_{p \in \Delta_m} \min_{q \in \Delta_n} p^\top A q = \min_{q \in \Delta_n} \max_{p \in \Delta_m} p^\top A q.}$$

Moreover, there are mixed strategies $p^* \in \Delta_m$, $q^* \in \Delta_n$ and a real number V such that

$$(p^*)^\top A q \geq V \quad \text{for every } q \in \Delta_n, \qquad p^\top A q^* \leq V \quad \text{for every } p \in \Delta_m.$$

Consequently, $(p^*)^\top A q^* = V$. The number V is called the *value of the game*.

Your goal is to prove this theorem using strong LP duality. Consider the following primal LP, whose variables are $p_1, \ldots, p_m$ and v:

$$\begin{aligned} \text{maximize} \quad & v \\ \text{subject to} \quad & v - \sum_{i=1}^{m} A_{ij} p_i \leq 0 \quad \text{for } j = 1, \ldots, n, \\ & \sum_{i=1}^{m} p_i = 1, \\ & p_i \geq 0 \quad \text{for } i = 1, \ldots, m, \\ & v \in \mathbb{R} \quad \text{(unrestricted in sign)}. \end{aligned}$$

1. For a fixed $p \in \Delta_m$, show that

$$\min_{q \in \Delta_n} p^\top A q = \min_{1 \leq j \leq n} \sum_{i=1}^{m} A_{ij} p_i.$$

   Use this to explain why the primal LP's optimal value equals Alice's largest guaranteed expected payoff. Show that the LP is feasible and its objective is bounded above.

First, the left side is at least as small as the right side, since for any j, we can always choose q to be the vector of all 0's except a 1 in position j and attain the same value.

Now, we need to show the other direction of the inequality.
We could prove this by convexity, since for a fixed $v = p^\top A$, vq is just a convex combination of the entries from v, and thus is always optimized at a vertex. For a more rigorous proof (since we didn't really talk about convex combinations in class):
Fix any q. We will show that $\min_{1\le j\le n} \sum_{i=1}^m A_{ij}p_i \le p^\top Aq$, and since this will hold for any q, it holds for the q that minimizes the left side, showing equality.

Since the entries of q sum to 1, it must contain nonzero entries. Let the indices of these nonzero entries be $s_1, ...s_k$. At least one of these entries is the "best", meaning it has the smallest value of $(p^\top A)_{s_i}$, which we can call s^*. Now,

$$\begin{aligned} p^\top Aq &= \sum_{i=1}^{n} (p^\top A)_i q_i \\ &= \sum_{i=1}^{k} (p^\top A)_{s_i} q_{s_i} \\ &\ge \sum_{i=1}^{k} (p^\top A)_{s^*} q_{s_i} \\ &= (p^\top A)_{s^*} \sum_{i=1}^{k} q_{s_i} \\ &= (p^\top A)_{s^*} \end{aligned}$$

Note the third step follows from the fact that s^* is "best" as we defined earlier and all entries of q are nonnegative. The last step follows from the fact that the nonzero entries of q must sum to 1.

That concludes the proof of the first part.

Now, let $v, p_1, ..., p_m$ be an optimal assignment to the LP. Note, we will have $v = \min_j \sum_{i=1}^m A_{ij}p_i$, since if not, we can increase v until it is, improving the reward and not breaking any constraints.
So, we also have $v = \min_q p^\top Aq$. Now, we maximize v. Thus, the LP is equivalent to taking $\max_p \min_q p^\top Aq$ for any valid choices of p, q, which is exactly Alice's largest guaranteed expected payoff.

The LP is feasible since we can set $p_1 = 1$ and $p_i = 0$ for all $i \neq 1$, and then set v to $\min_j A_{1j}p_1$, and we see that none of the constraints are violated. Its objective is bounded above since $\sum_{i=1}^m A_{ij}p_i \le \sum_{i=1}^m |A_{ij}|$, so v (the objective value) is bounded above by $\sum_{i=1}^m |A_{ij}|$ for any j.

2. Derive the dual LP, using $q_1, \ldots, q_n$ for the dual variables corresponding to the payoff constraints and w for the variable corresponding to $\sum_i p_i = 1$. State all sign restrictions.

   Hint: In a maximization primal, an equality constraint gives an unrestricted dual variable, and an

unrestricted primal variable gives an equality constraint in the dual.

We associate variables q_j with each of the constraints of the form $v - \sum ....$ We will associate variable w with the final constraint. This gives $n+1$ variables total.
Writing the dual, we get:

$$\begin{aligned}
&\min\ w \\
&s.t.\ w - \sum_{j=1}^{n} A_{ij} q_j \geq 0 && \text{for } i \in [m] \text{ } (m \text{ constraints assicated with } p_i\text{'s}) \\
&\sum_{i=1}^{n} q_i = 1 && (\text{constraint assicated with } v) \\
&q_i \geq 0, w \in \mathbb{R} \text{ (unrestricted in sign)}
\end{aligned}$$

3. Explain why the dual LP's optimal value equals Bob's smallest guaranteed upper bound on Alice's expected payoff. Apply strong LP duality to prove the displayed minimax equality. Finally, use optimal primal and dual solutions to obtain p^*, q^*, V satisfying both guarantees in the theorem.

Note the dual has essentially the same structure as the primal. If we replace w with $-w$ and replace A with $-A^\top$, we are actually in the exact same situation (call this the modified dual). So, we can use the same proof from part 1 to show that

$$\begin{aligned}
\min_{dualLP} w = - \max_{modifiedDualLP} \{-w\} &= - \max_{q} \min_{p} q^\top \left(-A^\top\right) p \\
&= - \max_{q} \left\{ - \max_{p} q^\top A^\top p \right\} \\
&= - - \min_{q} \max_{p} q^\top A^\top p \\
&= \min_{q} \max_{p} \left(q^\top A^\top p\right)^\top \\
&= \min_{q} \max_{p} p^\top A q
\end{aligned}$$

where the second to last step follows because the transposition of a scalar is itself. This proves the first part of the problem.

Now, by strong duality, them optima of the two LP's are are equal. This means $opt_{primal} = opt_{dual}$, meaning:

$$\max_{p \in \Delta_m} \min_{q \in \Delta_n} p^\top A q = \min_{q \in \Delta_n} \max_{p \in \Delta_m} p^\top A q.$$

which is exactly the minimax equality.

If we have optimal solutions $(p_1^*, ..., p_m^*, v^*)$ and $(q_1^*, ..., q_n^*, w^*)$, we see that $v^* = w^*$, which we can set to V, and we can define $p^* = (p_1^*, ..., p_m^*), q^* = (q_1^*, ..., q_n^*)$.

14 Convex Sets and Convex Functions

For each part, give a proof or a counterexample. You may not assume that convex functions are differentiable.

1. If $K, L \subseteq \mathbb{R}^n$ are convex sets, must $K \cap L$ be convex?

   $K \cap L$ is convex. This is because for any $x, y \in K \cap L$, we know $x, y \in K$ and $x, y \in L$. So, the segment $\overline{xy} \subset K$ by convexity of K, and $\overline{xy} \subset L$ by convexity of L.
   So, $\overline{xy} \subset K \cap L$. Since this is true for any such x, y, $K \cap L$ is convex.

   Must $K \cup L$ be convex?

   No. Let $K = \{0\}$ and $L = \{1\}$ in $\mathbb{R}^1$. Both sets are clearly convex, but the union contains both 0 and 1 and no other elements of the segment between them.

2. If $f, g : \mathbb{R} \to \mathbb{R}$ are convex functions, must $f + g$ be convex?

   Yes. Let $h = f + g$ Fix any $x, y \in \mathbb{R}$, $\lambda \in [0, 1]$. Now,

   $$\begin{aligned} h(\lambda x + (1-\lambda)y) &= f(\lambda x + (1-\lambda)y) + g(\lambda x + (1-\lambda)y) \\ &\leq \lambda f(x) + (1-\lambda)f(y) + \lambda g(x) + (1-\lambda)g(y) \\ &= \lambda h(x) + (1-\lambda)h(y) \end{aligned}$$

   Where the middle step follows from convexity of f and g.
   So, $h = f + g$ is convex.

3. If $f, g : \mathbb{R} \to \mathbb{R}$ are convex functions, must their pointwise product $h(x) = f(x)g(x)$ be convex?

   No. Let $f(x) = x^2$, $g(x) = -1$. Then, $h(x) = -x^2$ which is not convex.

4. If $f, g : \mathbb{R} \to \mathbb{R}$ are convex functions, must $h(x) = \max\{f(x), g(x)\}$ be convex?

   Yes Fix any $x, y \in \mathbb{R}$, $\lambda \in [0, 1]$. Now,

   $$\begin{aligned} h(\lambda x + (1-\lambda)y) &= \max\{f(\lambda x + (1-\lambda)y), g(\lambda x + (1-\lambda)y)\} \\ &\leq \max\{\lambda f(x) + (1-\lambda)f(y), \lambda g(x) + (1-\lambda)g(y)\} \\ &\leq \max\{\lambda f(x), \lambda g(x)\} + \max\{(1-\lambda)f(y), (1-\lambda)g(y)\} \\ &= \lambda \max\{f(x), g(x)\} + (1-\lambda)\max\{f(y), g(y)\} \\ &= \lambda h(x) + (1-\lambda)h(y) \end{aligned}$$

   The second step follows from convexity of f and g.
   The third step is true by looking at cases, $\max\{a+b, c+d\} \leq \max\{a, c\} + \max\{b, d\}$. Splitting up the max only gives more ways to combine the variables, which can only increase the max.
   So, h is convex.

15 Gradient Descent for Strongly-Convex Functions

A differentiable function f is α-strongly convex for $\alpha > 0$ if for all $x, y \in \mathbb{R}^n$ we have

$$f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle + \frac{\alpha}{2}\|y - x\|_2^2.$$

Consider an α-strongly convex differentiable function f with the 2-norm of its gradient always bounded by G. The goal is to minimize f and let x^* denote its minimum.

Show that the gradient descent algorithm with step size $\frac{1}{\alpha(t+1)}$ satisfies

$$f\Big(\frac{\sum_t x_t}{T}\Big) - f(x^*) \leq \frac{G^2(1+\log T)}{2\alpha T}.$$

Thus, strong-convexity allows us to get $1/T$ dependency in regret instead of $1/\sqrt{T}$ dependency for general convex functions.

(Hint: Change the potential function in the analysis from class to $\Phi(t) = \frac{t\alpha}{2}\|x_t - x^*\|^2$. Also, use that $\sum_{t\in\{1,\ldots,T\}} \frac{1}{t} \leq 1 + \log T$.)

Let n be the step size and f be our feedback function. Consider the potential function $\Phi(t) = \frac{\alpha t}{2}||x_t - x^*||^2$. Note that $||y_{t+1} - x^*||^2 \geq ||x_{t+1} - x^*||^2$ by the lecture notes (projection on K). Using this, as well as our step size $n = \frac{1}{\alpha(t+1)}$ and the upper bound on our gradient (G) along with the definition of α-strong convexity,

$$\Phi(t+1) = \frac{\alpha(t+1)}{2}(||x_{t+1} - x^*||^2) \leq \frac{\alpha(t+1)}{2}(||y_{t+1} - x^*||^2) = \frac{\alpha(t+1)}{2}||x_t - x^* - n\nabla f(x_t)||^2$$

$$= \frac{\alpha(t+1)}{2}\left(||x_t - x^*||^2 + (n\nabla f(x_t))^2 - 2n\langle x_t - x^*, \nabla f(x_t)\rangle\right)$$

$$= \Phi(t) + \frac{\alpha}{2}||x_t - x^*||^2 + \frac{(\nabla f(x_t))^2}{2\alpha(t+1)} - \langle x_t - x^*, \nabla f(x_t)\rangle$$

$$\leq \Phi(t) + f(x^*) - f(x_t) + \frac{G^2}{2\alpha(t+1)}$$

$$\implies f(x_t) - f(x^*) \leq \Phi(t) - \Phi(t+1) + \frac{G^2}{2\alpha(t+1)}$$

$$\implies \sum_t f(x_t) - f(x^*) \leq \Phi(0) - \Phi(T+1) + \frac{G^2}{2\alpha}\sum_t \frac{1}{t+1} \leq \frac{G^2(1+\log(T))}{2\alpha}$$

where the last step utilizes the observations that $\Phi(0) - \Phi(t+1) \leq 0$, and $\sum_t \frac{1}{t} \leq 1 + \log(T)$. Applying convexity to the left hand side, we have $f(\frac{\sum_t x_t}{T}) - f(x^*) \leq \frac{G^2(1+\log(T))}{2\alpha T}$ and we are done.